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PacBio HiFi Sequencing Powers First Arab Human Pangenome, Published in Nature Communications

Study reveals millions of previously unknown variants and genomic sequences from underrepresented populations

MENLO PARK, Calif., July 24, 2025 (GLOBE NEWSWIRE) — PacBio (NASDAQ: PACB), a leading developer of high-quality, highly accurate sequencing solutions, today announced a significant advancement for global genomic equity. Scientists have published the first Arab human pangenome in [Nature](#)



assembled high-resolution genomes from 53 individuals representing diverse Arab populations across eight countries.

This study results in millions of newly identified variants and over 111 million base pairs of previously unsequenced DNA, in genomic regions that were missing from widely used human references.

“MBRU’s work demonstrates why population-specific pangenomes matter,” said Christian Henry, President and Chief Executive Officer of PacBio. “We are proud that MBRU, one of our earliest Revio customers, has delivered such a historic contribution to global genomics. This study will have lasting impact for research and precision medicine in a historically underrepresented population.”

Despite comprising nearly 6% of the global population, Arab individuals have been largely absent from the datasets that inform genomic research and medicine. Using PacBio HiFi sequencing in combination with other long-read and ultra-long-read technologies, scientists assembled diploid, haplotype-resolved genomes at exceptional quality, uncovering:

- 111 million base pairs of uncharacterized human genome sequences absent from existing human pangenomes, the T2T-CHM13 and GRCh38 reference human genomes and large-scale databases.
- 8.94 million small and 235,000 structural variants unique to Arab individuals, many undetectable by short-read methods.
- 883 duplicated genes, including *TAF11L5*, found in every individual studied and potentially linked to recessive disease.
- Over 1,400 base pairs of new mitochondrial DNA, adding resolution to maternal lineage tracking in Arab populations.



“By creating a high-resolution Arab pangenome we’re giving scientists and clinicians a tool to make precision medicine more equitable,” said Dr. Mohammed Uddin, senior author of the study and associate professor of human genetics at MBRU. “This achievement depended on long-read technologies, notably PacBio’s high quality HiFi sequencing, which allowed us to capture the full complexity of the genome, including structural variation and previously hidden sequences.”

Compared to existing references, the Arab pangenome, or UAE Pangenome Reference (UPR) delivered improved mapping rates, better variant recall, and greater insight into regions tied to recessive disease, many of which are more prevalent in Arab populations due to genetic structure and inheritance patterns.

It is freely available and intended to support ongoing research into genetic disorders, disease predisposition, and population health. Early data suggests that using the UPR as a reference can significantly increase variant detection accuracy in Arab genomes, helping scientists and clinicians better interpret genetic data from this population.

The publication follows other major pangenome initiatives supported by PacBio, including contributions to the Human Pangenome Reference Consortium (HPRC) and the Chinese Pangenome Consortium (CPC), further affirming PacBio’s leadership in global genomics.

About PacBio

PacBio (NASDAQ: PACB) is a premier life science technology company that designs, develops, and manufactures advanced sequencing solutions to help scientists and clinical researchers resolve genetically complex problems. Our products and



infectious disease and microbiology, oncology, and other emerging applications. For more information, please visit www.pacb.com and follow @PacBio.

PacBio products are provided for Research Use Only. Not for use in diagnostic procedures.

Forward-Looking Statements

This press release contains “forward-looking statements” within the meaning of Section 21E of the Securities Exchange Act of 1934, as amended, and the U.S. Private Securities Litigation Reform Act of 1995. All statements other than statements of historical fact are forward-looking statements, including statements relating to the uses, advantages, quality or performance of, or the benefits or expected benefits of using, PacBio products or technologies, including in connection with the Arab human pangenome publication and its anticipated impact on medical research, precision medicine, and increased variant detection, and other future events. You should not place undue reliance on forward-looking statements because they are subject to assumptions, risks, and uncertainties that could cause actual outcomes and results to differ materially from currently anticipated results. These risks include, but are not limited to, rapidly changing technologies and extensive competition in genomic sequencing; unanticipated increases in costs or expenses; and other risks associated with general macroeconomic conditions and geopolitical instability. Additional factors that could materially affect actual results can be found in PacBio’s most recent filings with the Securities and Exchange Commission, including PacBio’s most recent reports on Forms 8-K, 10-K, and 10-Q, and include those listed under the caption “Risk Factors.” These forward-looking statements are based on current expectations and speak only as of the date hereof; except as



available.

Contacts (MBRU)

Eva Davies

eva.davies@dubaihealth.ae

Contacts (PacBio)

Investors:

Todd Friedman

ir@pacificbiosciences.com

Media:

pr@pacificbiosciences.com

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